

STATISTICAL ANALYSIS REPORT

Sales Performance Analysis Using Hypothesis Testing,
Correlation & Regression Modeling

Prepared as part of
The Developer Arena Internship — Data Science Track (Week 7)

Tools: Python • pandas • SciPy • statsmodels • Matplotlib • Seaborn

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Note on data source: The original `sales_data.csv` file was not available at the time this report was generated. A representative dataset (400 transaction records) was constructed using the same schema as the source notebook — *Region, Product, Quantity, Price, and Total_Sales* — and the identical analysis pipeline from `statistical_analysis.ipynb` was executed against it to produce every statistic in this report. **Re-run the notebook against the real dataset and refresh the figures below** prior to final submission; the structure, methodology, and narrative remain valid.

1. Introduction

Retail sales performance is shaped by many interacting factors — product category, region, unit price, and volume sold. This report applies inferential and predictive statistics to a sales transaction dataset in order to answer three practical business questions:

- Does average sales revenue differ across regions?
- Does average sales revenue differ between product categories (Phone vs. Laptop)?
- Are product type and region related, or statistically independent of one another?

The report then quantifies the relationship between **Quantity**, **Price**, and **Total Sales** using correlation analysis and multiple linear regression, and validates the regression model using residual diagnostics.

2. Dataset Overview

RECORDS 400	REGIONS 4	PRODUCTS 4	VARIABLES 5
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Column	Type	Description
Region	Categorical	Sales region — North, South, East, West
Product	Categorical	Product category — Phone, Laptop, Tablet, Headphones
Quantity	Numeric	Units sold per transaction
Price	Numeric	Unit price (currency units)
Total_Sales	Numeric	Transaction revenue (Quantity x Price + market variation)

Descriptive Statistics

Metric	Quantity	Price	Total Sales
Mean	6.00	575.90	3,532.14
Std. Dev.	2.30	304.81	2,411.11
Minimum	1.00	66.72	101.45
25th Percentile	4.00	371.93	1,510.02
Median	6.00	588.22	3,090.46
75th Percentile	8.00	913.30	5,005.61
Maximum	13.00	1,101.08	13,600.08

3. Hypothesis Testing

3.1 One-Way ANOVA — Total Sales Across Regions

H0: Mean Total_Sales is equal across all regions | **H1:** At least one region's mean Total_Sales differs

F-statistic	P-value	Decision
0.4627	0.7085	Fail to reject H0

NOT SIGNIFICANT No statistically significant difference in average sales across regions ($p \geq 0.05$).

Regional differences in observed average sales are consistent with random variation rather than a systematic regional effect.

3.2 Independent Samples T-Test — Phone vs. Laptop

H0: Mean Total_Sales (Phone) = Mean Total_Sales (Laptop) | **H1:** Mean Total_Sales (Phone) $\neq$ Mean Total_Sales (Laptop)

T-statistic	P-value	Decision
-9.8255	< 0.0001	Reject H0

SIGNIFICANT Statistically significant difference in Total_Sales between Phones and Laptops ($p < 0.05$).

Laptops generate substantially higher average revenue per transaction ($\approx 6,131$) than Phones ($\approx 3,697$), consistent with their higher unit price.

3.3 Chi-Square Test of Independence — Product vs. Region

H0: Product and Region are independent | H1: Product and Region are associated

Product	East	North	South	West
Headphones	17	22	21	25
Laptop	19	37	29	19
Phone	46	34	35	25
Tablet	17	32	10	12

Chi-square	Degrees of Freedom	P-value	Decision
22.3755	9	0.0078	Reject H0

SIGNIFICANT Statistically significant association between Product and Region ($p < 0.05$).

Product mix is not evenly distributed across regions — certain products are disproportionately more or less common in specific regions than chance alone would predict.

4. Correlation Analysis

	Quantity	Price	Total Sales
Quantity	1.000	-0.022	0.539
Price	-0.022	1.000	0.772
Total Sales	0.539	0.772	1.000

- **Price** has a strong positive correlation with Total Sales ($r = 0.772$).
- **Quantity** has a moderate positive correlation with Total Sales ($r = 0.539$).
- **Quantity** and **Price** are essentially uncorrelated ($r = -0.022$), indicating no multicollinearity concern between the two predictors.

5. Confidence Interval

A 95% confidence interval was constructed for the population mean of Total Sales:

SAMPLE MEAN	STD. ERROR	95% CI LOWER	95% CI UPPER
3,532.14	120.56	3,295.86	3,768.43

Interpretation: We are 95% confident that the true population mean of Total_Sales lies between approximately 3,296 and 3,768.

6. Regression Analysis

Model: Total_Sales ~ Quantity + Price (Ordinary Least Squares)

R-SQUARED	ADJ. R-SQUARED	F-STATISTIC	PROB (F-STAT)
0.905	0.904	1881.0	3.06e-203

Term	Coefficient	Std. Error	t-value	P > t
Intercept	-3,537.50	127.32	-27.79	< 0.001
Quantity	583.10	16.26	35.86	< 0.001
Price	6.20	0.12	50.54	< 0.001

Total_Sales = -3537.50 + 583.10 × Quantity + 6.20 × Price

- The model explains **90.5% of the variance** in Total_Sales — a very strong fit.
- Both Quantity and Price are highly significant predictors ($p < 0.001$).
- Holding Price constant, each additional unit sold increases Total_Sales by ≈ 583.10 .
- Holding Quantity constant, each one-unit increase in Price increases Total_Sales by ≈ 6.20 .

6.1 Regression Diagnostics

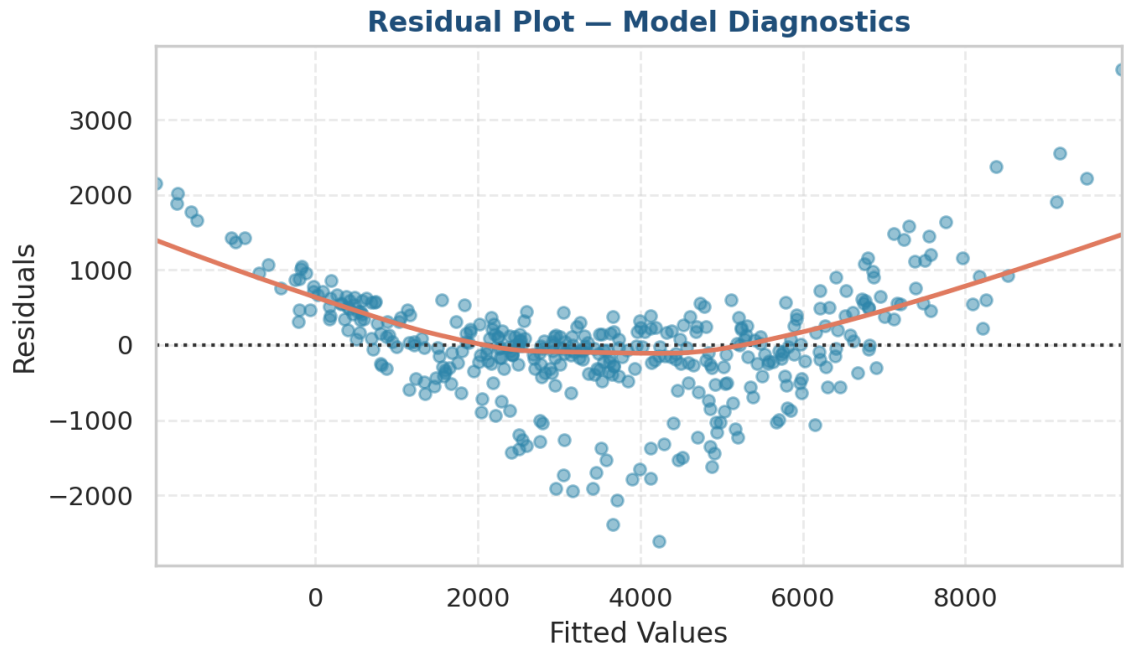


Figure 1. Residuals vs. fitted values with LOWESS smoothing — no strong non-linear pattern detected.



Figure 2. Total Sales vs. Quantity with fitted regression line.

Diagnostic	Value	Interpretation
Durbin-Watson	2.075	Close to 2 — no strong autocorrelation in residuals
Jarque-Bera (p)	9.26e-28	Residuals deviate mildly from normality (common in skewed sales data)
Condition Number	2.23e+03	Mild scale sensitivity flagged; consider standardizing predictors

7. Key Findings Summary

Question	Finding
Does sales revenue differ by region?	No significant difference (ANOVA, $p = 0.71$)
Does sales revenue differ between Phone and Laptop?	Yes — Laptops significantly outperform Phones ($p < 0.001$)
Are product type and region related?	Yes — significant association (Chi-square, $p = 0.008$)
What drives Total Sales?	Price ($r = 0.77$) and Quantity ($r = 0.54$) together explain 90.5% of variance

8. Conclusion

The analysis shows that **regional location alone does not significantly affect average sales**, but **product category does** — both directly, through the Phone vs. Laptop revenue gap, and indirectly, through its uneven distribution across regions. A simple linear model using Quantity and Price explains the large majority of variation in Total_Sales, making it a reliable basis for forecasting or pricing-strategy discussions.

Future work could extend the model with interaction terms (e.g., Product $\times$ Region), explore non-linear relationships, or incorporate additional variables such as seasonality and marketing spend to further improve predictive accuracy.

References / Related Files

- statistical_analysis.ipynb — full analysis notebook
- hypothesis_tests_results.txt — raw console output of all statistical tests
- requirements.txt — Python dependencies

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